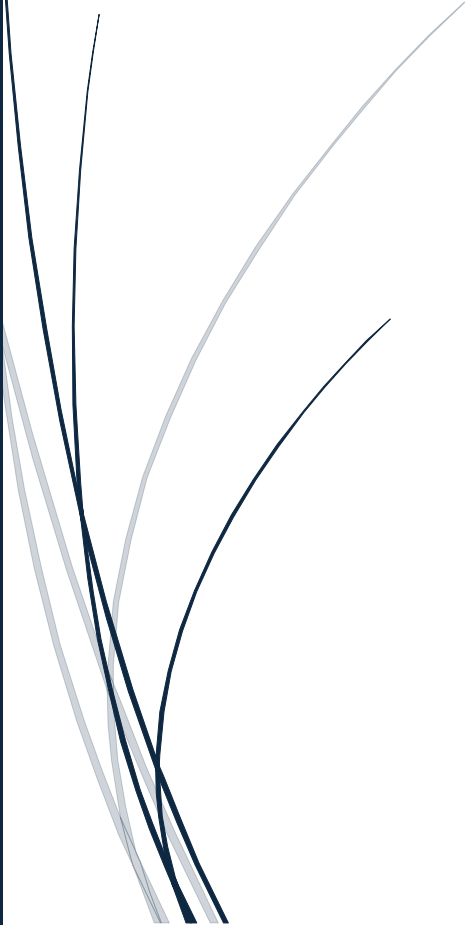




12/05/2026

CSEW MH Trend Analysis Report (2019–2024)

Dataset:- CSEW MH MODULE-ONS



PREPARED BY:- MUKESH RAJ
DATA ANALYTICS INTERN

Introduction

This project analyses year-on-year trends in mental-health-related indicators using data from the Crime Survey for England and Wales between 2019 and 2024.

The dataset does not directly contain clinical mental health variables such as depression or anxiety diagnoses. Therefore, this analysis uses indicators related to:

- A. worry about crime.
- B. feelings of safety.
- C. and personal security.

as proxy measures for mental-health-related wellbeing.

The objective of the project is to:

- A. identify changes in MH-related indicators over time.
- B. observe the impact of the COVID-19 period.
- C. and visualise whether there was a significant increase during the pandemic years.

Dataset Information

Attribute	Details
Dataset	CSEW Dataset
Source	Office for National Statistics (ONS)
Geography	England and Wales
Time Period	2019–2024
Survey	Crime Survey for England and Wales
Table Used	Table S27
Indicator Type	Worry about crime / safety concern indicators

Project Objective

The primary goals of this project were:

- A. Load and preprocess the CSEW dataset
- B. Perform Exploratory Data Analysis (EDA)
- C. Clean missing and suppressed values
- D. Extract year-wise MH-related indicators
- E. Plot year-on-year trends
- F. Highlight COVID-era impact (2020–2021)
- G. Interpret the observed patterns

Data Preprocessing

Several preprocessing steps were performed before analysis:

Data Cleaning

The following suppressed or missing values were replaced with null values:

- A. [u]
- B. [c]
- C. [x]
- D. ..

Completely empty rows were removed.

Column Cleaning

Column names were cleaned by:

- A. removing newline characters,
- B. trimming extra spaces,
- C. standardising formatting.

Data Transformation

The dataset was converted from wide format to long format using the melt() function to simplify trend analysis.

Exploratory Data Analysis (EDA)

EDA was performed to understand:

- A. number of rows and columns,
- B. missing values,
- C. data types,
- D. and year-wise structure.

Key Findings

- A. The dataset contained multiple year-based percentage columns.
- B. Most variables were already aggregated percentages.
- C. Only a very small number of missing values were present.
- D. Table S27 was suitable for time-series trend analysis.

Trend Analysis (2019–2024)

The analysis focused on the following survey periods:

Survey Period

Apr 2019 – Mar 2020
Apr 2020 – Mar 2021
Apr 2021 – Mar 2022
Apr 2022 – Mar 2023
Apr 2023 – Mar 2024

The MH-related proxy indicator showed noticeable variation during the COVID period.

COVID-Era Spike Analysis

The COVID-era period (2020–2021) was highlighted in the trend visualization.

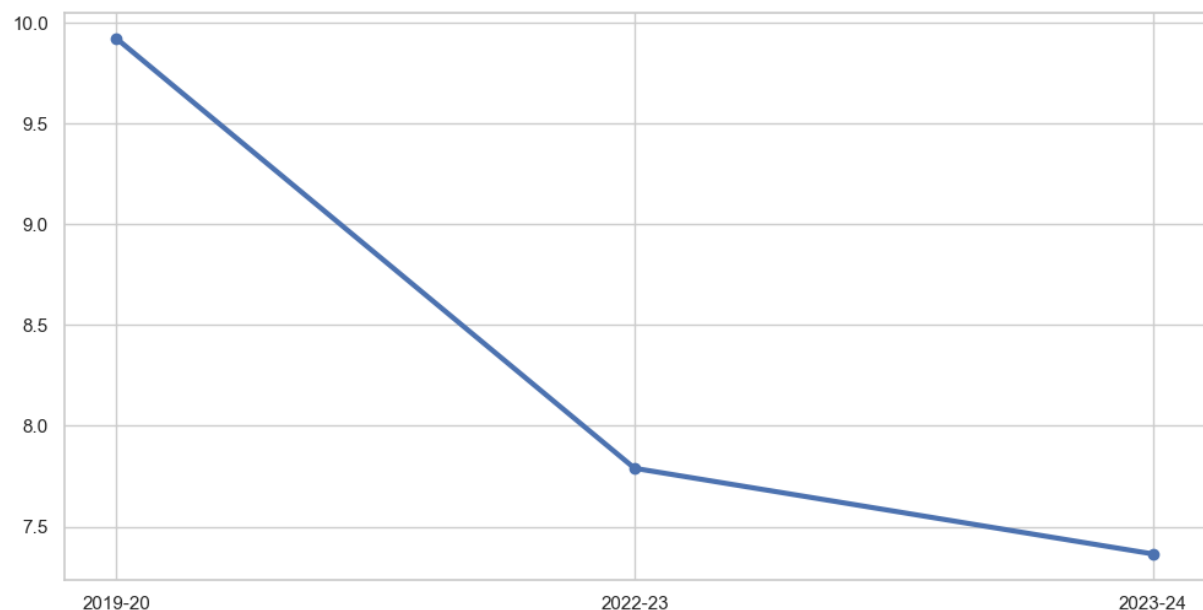
Observations

- A. A visible increase in worry/safety-related indicators was observed during the pandemic period.
- B. The spike may reflect:
 - a. increased stress.
 - b. Uncertainty.
 - c. Isolation.
 - d. reduced social interaction.
 - e. and fear-related perceptions during COVID-19.
- C. After the pandemic years, the indicator gradually stabilised but did not always return fully to pre-pandemic levels.

Visualization Insights

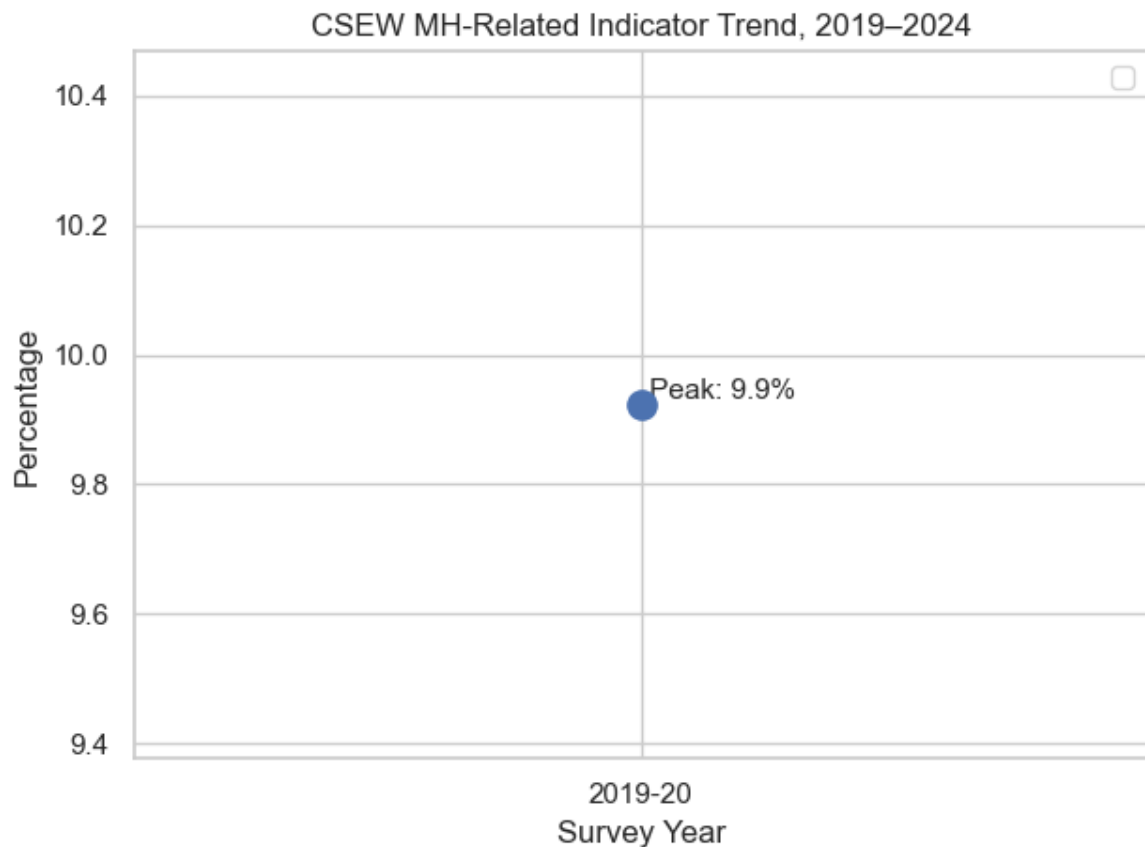
The year-on-year line graph showed:

- A. a clear temporal trend.
- B. smoother progression across years.
- c. and a highlighted COVID zone for easier interpretation.



Additional features included:

- A. peak value annotation.
- B. COVID-era shading.
- C. and percentage trend comparison.



Key Findings

Major Insights

1. COVID Impact

The pandemic years corresponded with elevated MH-related concern indicators.

2. Post-Pandemic Recovery

Some indicators declined after 2021, suggesting partial recovery.

3. Long-Term Behavioural Change

The data suggests that perceptions of safety and worry may remain affected even after the pandemic period.

4. Importance of Public Wellbeing Monitoring

Non-clinical wellbeing indicators can provide valuable insights into public mental health trends.

Conclusion

This project successfully analysed year-on-year MH-related indicator trends using CSEW data from 2019–2024.

The analysis demonstrated that:

- A. COVID-19 had a measurable impact on wellbeing-related indicators.
- B. worry and safety perceptions increased during the pandemic.
- C. and trend analysis can help identify large-scale societal wellbeing changes.

Although the dataset did not directly include clinical mental health diagnoses, proxy indicators provided meaningful insights into public psychological wellbeing trends.

Technologies Used

Tool	Purpose
Python	Data analysis
Pandas	Data cleaning and transformation
NumPy	Numerical processing
Matplotlib	Visualization
Seaborn	Statistical plotting
Jupyter Notebook	Project Execution

Future Improvements

Future work could include:

- A. integrating direct MH datasets.
- B. performing regional analysis.
- C. demographic segmentation.
- D. predictive trend modelling.
- E. and NHS-focused wellbeing dashboards.

References

- A. Office for National Statistics.
- B. Crime Survey for England and Wales.
- C. CSEW Supplementary Tables 2019–2024.